AI-Enhanced Sprint Planning & Execution

Lecture 1: AI-Assisted Prioritization and Estimation

Step 1: Apply MoSCoW Prioritization and Story Point Estimation

Input Documents Required:

- `user_stories_backlog.csv` (Story ID, Title, Description, Business Value, Dependencies)
- `reference_stories.csv` (previously estimated stories with story points)

COSTAR Prompt:

CONTEXT: You are an Agile Product Owner working with a development team on sprint planning. We have user stories in the backlog that need to be prioritized and estimated. The team uses the Scrum framework and Fibonacci sequence (1, 2, 3, 5, 8, 13, 21) for story points.

OBJECTIVE: First, apply the MoSCoW prioritization method (Must have, Should have, Could have, Won't have) to categorize each story based on business value and dependencies. Then, suggest story point estimates by comparing new stories against the reference stories provided.

STYLE: Analytical and structured, following Agile best practices.

TONE: Professional and objective with clear reasoning.

AUDIENCE: Product Owner, Scrum Master, and Development Team in planning session.

RESPONSE: Provide a comprehensive table with columns: Story ID, Title, MoSCoW Category, Suggested Story Points, Comparison Reference, Rationale. Include a summary showing: (1) Count of stories per MoSCoW category, (2) Total story points for Must/Should categories, (3) Any dependencies or risks flagged.

Step 2: Identify Issues and Refinement Needs

Input Documents Required:

- `prioritized_estimated_backlog.csv` (output from Step 1)

COSTAR Prompt:

CONTEXT: You are a Scrum Master reviewing the prioritized and estimated backlog before finalizing sprint planning. You need to ensure the backlog is ready and identify any stories that need refinement.

OBJECTIVE: Analyze the backlog to identify: (1) Stories with unusually high story points (13+) that should be split, (2) Stories with unclear requirements or acceptance criteria, (3) Stories with high-risk dependencies, (4) Estimation inconsistencies when compared to similar stories.
Refer to the estimates and initially uploaded backlog.

STYLE: Quality-focused and constructive, using Agile refinement criteria.

TONE: Helpful and improvement-oriented.

AUDIENCE: Scrum Master and Product Owner preparing for final sprint planning.

RESPONSE: Provide a prioritized list organized by severity (High Priority, Medium Priority, Low Priority). For each flagged story include: Story ID, Issue Type, Specific Concern, Recommended Action (e.g., "Split into smaller stories", "Schedule refinement session", "Clarify acceptance criteria").

Lecture 2: Sprint Planning

Step 1: Calculate Team Capacity and Generate Sprint Backlog

Input Documents Required:

- `team_availability.csv` (Team member names, available hours per day, planned absences)
- `historical_velocity.csv` (last 3-5 sprints' velocity)
- `prioritized_estimated_backlog.csv`
- `sprint_parameters.txt` (Sprint length, dates)

COSTAR Prompt:

CONTEXT: You are planning a 2-week sprint for a Scrum team of 6 developers. The team has varying availability due to vacations and meetings. The team's focus factor is 70% (time actually spent on sprint work). Historical data shows the team averages 60-70 story points per sprint.

OBJECTIVE: First, calculate total team capacity in story points considering individual availability and focus factor. Then, recommend a sprint backlog by selecting stories that: (1) Fit within capacity, (2) Prioritize "Must Have" items, (3) Include 10% buffer for unexpected work, (4) Form a coherent sprint goal.

STYLE: Quantitative and strategic, showing clear calculations.

TONE: Recommendatory and transparent.

AUDIENCE: Product Owner and Development Team during sprint planning ceremony.

RESPONSE: Provide:
1. **Capacity Calculation Table** (Team member, Available hours, Adjusted capacity)
2. **Total Sprint Capacity** (in story points with calculation shown)
3. **Recommended Sprint Backlog Table** (Story ID, Title, Story Points, Priority, Total: 90% of capacity)

4. **Proposed Sprint Goal** (one clear sentence)
5. **Key Dependencies** and risks to watch

Step 2: Create Sprint Plan with Sequencing

Input Documents Required:

- `approved_sprint_backlog.csv` (stories selected by team)
- `sprint_goal.txt`

COSTAR Prompt:

CONTEXT: You are documenting the sprint planning outcomes. The team has finalized their sprint backlog and agreed on a sprint goal. Some stories have dependencies that affect execution sequence.

OBJECTIVE: Create a comprehensive Sprint Plan document that includes: (1) Sprint overview and goal, (2) Complete backlog with story sequencing based on dependencies, (3) Recommended implementation timeline (Week 1 vs Week 2), (4) Critical path stories, (5) Success metrics and key dates.

STYLE: Formal documentation, well-organized and actionable.

TONE: Clear and authoritative, suitable for team reference.

AUDIENCE: Development Team (daily reference), Stakeholders (transparency), Scrum Master.

RESPONSE: Create a structured Sprint Plan with these sections:
1. **Sprint Overview** (Number, dates, goal statement)
2. **Sprint Backlog Table** (Story ID, Title, Story Points, Week, Priority, Dependencies)
3. **Execution Sequence** (which stories to start Day 1-3, Day 4-7, Day 8-10)
4. **Critical Path Stories** (must complete early to unblock others)
5. **Sprint Success Criteria** (Definition of Done, velocity target)
6. **Key Dates** (Daily stand-up time, Sprint Review, Retrospective)

Lecture 3: Identifying and Mitigating Impediments

Step 1: Analyze Stand-ups and Identify Impediments

Input Documents Required:

- `daily_standup_notes.txt` (5-7 days of stand-up updates)
- `sprint_progress_data.csv` (Story ID, Status, Days in Status, Time Spent) - ***create Sample***

COSTAR Prompt:

CONTEXT: You are a Scrum Master analyzing the first week of sprint execution. Team members have been providing daily stand-up updates (what they did, what they'll do, blockers). You also have data on story progress. You need to identify both explicit blockers and implicit issues.

OBJECTIVE: Analyze the stand-up notes and progress data to identify: (1) Explicit impediments mentioned by team members, (2) Implicit problems (stories stuck in status, repeated mentions without progress), (3) Patterns indicating systemic issues, (4) Team velocity risks, (5) Predict potential future blockers before they escalate.

STYLE: Analytical and pattern-recognition focused, like a detective.

TONE: Observant, non-judgmental, and proactive.

AUDIENCE: Scrum Master who will take action and Development Team.

RESPONSE: Provide a structured **Impediment Analysis Report** with:
1. **Active Impediments Table** (Team Member, Date Raised, Blocker Description, Severity: High/Medium/Low, Days Open)
2. **Pattern Analysis** (recurring themes, knowledge silos, dependency delays)
3. **At-Risk Stories** (stories showing warning signs: Story ID, Risk Indicator, Days Stuck)
4. **Predictive Alerts** (potential future blockers based on trends)
5. **Priority Action Items** (what needs immediate attention)

Step 2: Generate Resolution Strategies

Input Documents Required:

- `active_impediments.csv` (from Step 1 analysis)
- `team_skills_matrix.csv` (team member skills and expertise)
- `stakeholder_contacts.csv` (external contacts for escalation)

COSTAR Prompt:

CONTEXT: You are an experienced Agile Coach helping resolve impediments identified during the sprint. The team has several active blockers affecting sprint progress. You have information about team skills and external stakeholders who can help.

OBJECTIVE: For each active impediment, generate practical resolution strategies. Consider: (1) Internal solutions (re-pairing, skill-sharing, workarounds), (2) External escalations (specific people to contact), (3) Immediate vs. long-term fixes, (4) Time-to-resolve estimates, (5) Prevention measures for the future.

STYLE: Solution-oriented and actionable, with clear steps.

TONE: Empowering and resourceful, helping team self-organize.

AUDIENCE: Scrum Master (facilitating resolution) and Development Team.

RESPONSE: For each impediment, provide:
1. **Impediment Summary** (brief description and impact)
2. **Resolution Strategy Table** (Option, Actions Required, Owner, Time Estimate, Pros/Cons)
3. **Recommended Approach** (best option with rationale)
4. **Escalation Path** (if strategy fails, who to escalate to)
5. **Prevention Measures** (how to avoid this in future sprints)
Format as clear action plans the team can execute immediately.

Lecture 4: Burndown Planning and Tracking with AI

Step 1: Create Burndown Projection and Track Progress

Input Documents Required:

- `sprint_backlog_with_points.csv` (Story ID, Story Points)

COSTAR Prompt:

CONTEXT: You are managing a sprint with committed story points. At the sprint start, you need to create an ideal burndown projection. During the sprint (currently Day 6), you need to track actual progress against the ideal and identify any concerning trends.

OBJECTIVE:
- **At Sprint Start**: Generate ideal burndown plan showing day-by-day expected story points remaining.
- **At Day 6 (Mid-Sprint)**: Compare actual (create sample data in a column called "actual SP remaining, show some lag)
vs. ideal burndown, analyze velocity, and determine if sprint goal is at risk. Provide early warnings if the team is falling behind.

STYLE: Quantitative and data-driven with clear metrics.

TONE: Factual and objective, providing honest assessment.

AUDIENCE: Scrum Master and Development Team for daily tracking.

RESPONSE: Provide:
1. **Ideal Burndown Table** (Day, Date, Ideal Points Remaining, Daily Burn Rate)
2. **Current Status** (Day X: Points Burned/Remaining, Actual vs. Ideal variance)
3. **Velocity Analysis** (current daily average vs. required average to complete on time)
4. **Trend Indicator** (🟢 Green: on track | 🟡 Yellow: at risk | 🔴 Red: significant deviation)
5. **Forecast** (based on current velocity, will we meet sprint goal?)
6. **Recommendations** (continue pace / accelerate / consider descoping specific stories)

Step 2: Sprint Completion Forecast and Retrospective Analysis

Input Documents Required:

- `mid_sprint_burndown_data.csv` (Day 6-7 actual progress)
- `remaining_stories.csv` (incomplete stories with complexity)
- `final_burndown_data.csv` (complete sprint data - for retrospective)
- `sprint_events_log.txt` (impediments, scope changes during sprint)

COSTAR Prompt:

CONTEXT: You are on Day 7 of a 10-day sprint. The team needs a forward-looking forecast to decide on final actions. After sprint completion, you need retrospective insights from the burndown data.

OBJECTIVE:
- **Day 7 (Forecasting)**: Project sprint completion using multiple scenarios (optimistic/realistic/pessimistic). Identify at-risk stories and recommend final 3-day action plan.
- **Post-Sprint (Retrospective)**: Analyze complete burndown to extract learnings about estimation accuracy, velocity patterns, and improvement opportunities.

STYLE: Predictive for forecasting; reflective for retrospective. Data-driven throughout.

TONE: Realistic and actionable for forecasting; constructive and learning-focused for retrospective.

AUDIENCE: Product Owner (scope decisions) and Scrum Team (retrospective).

RESPONSE: Provide:

FOR FORECASTING (Day 7):
1. **Current Metrics** (45/62 points burned, 3 days remaining)
2. **Completion Scenarios** (Optimistic/Realistic/Pessimistic with probability %)
3. **At-Risk Stories Table** (Story ID, Points, Completion Probability)
4. **Scope Recommendation** (if needed: stories to defer to next sprint)
5. **Final 3-Day Action Plan** (specific daily targets and focus)

FOR RETROSPECTIVE (Post-Sprint):
1. **Sprint Outcome** (goal achieved? Planned vs. actual story points)
2. **Burndown Pattern Analysis** (front-loaded, back-loaded, or steady?)
3. **Key Deviations** (when and why we deviated from ideal)
4. **Estimation Accuracy** (were estimates realistic?)
5. **Improvement Suggestions** (3-5 data-driven recommendations for next sprint)

Summary of Input Documents

Lecture	Input Documents Needed
Lecture 1	• user_stories_backlog.csv • reference_stories.csv
Lecture 2	• team_availability.csv • historical_velocity.csv • sprint_parameters.txt • story_dependencies.csv
Lecture 3	• daily_standup_notes.txt • sprint_progress_data.csv • team_skills_matrix.csv • stakeholder_contacts.csv
Lecture 4	• sprint_backlog_with_points.csv • sprint_calendar.csv • daily_story_completion.csv • sprint_events_log.txt